🖥️ Lab Sheet — Class, Object, Constructor, Destructor, Setter & Getter in C#

🎯 Objectives

- Learn how to define a **class** and create **objects**.
- Understand **constructors** (for initialization) and **destructors** (for cleanup).
- Implement **setters and getters** for controlled access to private fields.

📖 Theory

- **Class**: Blueprint that defines properties and methods.
- **Object**: Instance of a class created using `new`.
- **Constructor**: Special method called when an object is created.
- **Destructor**: Special method called when an object is destroyed (rarely used in modern C# because of garbage collection).
- **Setter/Getter**: Methods or properties used to access private fields safely.

📝 Step-by-Step Guidelines

1. Create a new **Console Application** project.
2. Define a `Student` class with private fields.
3. Add a **constructor** to initialize values.
4. Add a **destructor** to show cleanup.
5. Implement **getters and setters** using properties.
6. Create objects and test functionality.

💻 Full Example Code

Student.cs

```
using System;

public class Student
{
    // Private fields
    private string name;
    private int age;

    // Constructor (called when object is created)
    public Student(string name, int age)
    {
        this.name = name;
        this.age = age;
        Console.WriteLine("Constructor called: Student object created.");
    }

    // Destructor (called when object is destroyed)
    ~Student()
    {
        Console.WriteLine("Destructor called: Student object destroyed.");
    }

    // Getter and Setter using properties
    public string Name
    {
        get { return name; }
        set { name = value; }
    }

    public int Age
    {
        get { return age; }
        set
        {
            if (value > 0)
                age = value;
            else
                Console.WriteLine("Age must be positive.");
        }
    }

    // Method to display info
    public void DisplayInfo()
    {
        Console.WriteLine($"Name: {Name}, Age: {Age}");
    }
}
```

Program.cs

```
using System;

class Program
{
    static void Main(string[] args)
    {
        // Create object using constructor
        Student s1 = new Student("Harun", 22);

        // Display info
        s1.DisplayInfo();

        // Use setter to update values
        s1.Name = "Ripa";
        s1.Age = 21;

        // Display updated info
        s1.DisplayInfo();

        // Destructor will be called automatically when program ends
        Console.ReadLine();
    }
}
```

📊 Expected Output

```
Constructor called: Student object created.
Name: Harun, Age: 22
Name: Ripa, Age: 21
Destructor called: Student object destroyed.
```

🧪 Lab Exercises

1. Add a **Department property** with getter/setter.
2. Prevent setting **Age below 18** (validation in setter).
3. Create multiple `Student` objects and store them in a **List**.
4. Add a method to calculate **average age** of all students.
5. Observe when the **destructor** is called (usually at program end).